



Experiment No. 2

TITLE: Data Collection from Social Media Platforms using Web Scraping, Crawling, and Parsing.

AIM: To collect and capture social media data for business applications using web scraping, crawling, and parsing techniques with Python.

THEORY:

Introduction to Data Collection in Social Media Analytics

Social Media Analytics focuses on extracting meaningful insights from digital platforms such as Facebook, Twitter, YouTube and LinkedIn. Data collection is the foundational step in social media analytics because all analysis, visualization, modelling, and dashboard development depend on the availability of structured and relevant data.

The collected data may consist of posts, comments, reviews, likes, shares, hashtags, and engagement statistics. Businesses use this information to understand customer behaviour, monitor brand reputation, analyse competitor strategies, and identify market trends.

Web Scraping

Web scraping is an automated technique used to extract structured information from websites. It involves sending a request to a webpage, retrieving its HTML content, parsing the structure, and extracting relevant information. In social media analytics, scraping enables the collection of publicly available data when official APIs are limited or restricted.

Scraping is especially useful for gathering blog comments, product reviews, trending discussions, and textual content that can later be analyzed for sentiment, topic modeling, or trend detection.

Key Techniques in Data Collection

a) Scraping

Scraping extracts specific information from web pages.

Example:

- Extracting reviews from an e-commerce website.
- Collecting blog comments for sentiment analysis.

b) Crawling

A crawler, also known as a spider, is a program that systematically browses multiple web pages by following hyperlinks. Crawling is generally used when data needs to be collected from multiple interconnected pages rather than a single webpage. **c) Parsing**

Parsing refers to analysing the HTML structure of a webpage and converting it into a structured format known as the Document Object Model. This structured representation allows the program to locate specific tags and extract required information efficiently. **Tools Used for Web Scraping a) Requests Library**

The Requests library in Python is used to establish communication with a website and retrieve its HTML source code through HTTP requests. It simplifies the process of connecting to a server and obtaining webpage content.

b) BeautifulSoup Library

The BeautifulSoup library is used for parsing HTML content. It allows navigation through tags, attributes, and elements within the webpage structure. By identifying specific tag names and class attributes, the required textual or structural data can be extracted accurately.

The Web Scraping Process

A typical scraping workflow consists of four stages:

1. **Connect to the Website** ○ Use `requests.get()` to fetch the target URL.
 - Always check the response status code (e.g., 200 for success).
 - Respect `robots.txt` and implement polite delays (`time.sleep()`) to avoid overloading the server.
2. **Parse the HTML Content** ○ Pass the HTML string to BeautifulSoup to obtain a navigable object representing the document tree (DOM).
3. **Extract Required Data** ○ Use methods like `find()`, `find_all()` to locate specific HTML tags (e.g., `<div>`, `<span>`) based on attributes like class or id.
 - Extract text using `.get_text()` or attribute values (e.g., href from an `<a>` tag).
4. **Store Data in a Structured Format** ○ Organise the extracted data into Python data structures (lists of dictionaries).
 - Write to a **CSV file** (using `csv.writer` or `pandas.DataFrame.to_csv`) or insert into **MongoDB** for later processing.

Applications in Social Media Analytics

The data collected through web scraping directly supports several SMA tasks outlined in the syllabus:

- **Sentiment Analysis** – Scraping product reviews or social media comments to classify opinions as positive, negative, or neutral (Experiment 5).
- **Trend Analysis** – Extracting hashtags, keywords, or mentions over time to detect emerging topics
- **Customer Feedback Analysis** – Collecting reviews from e-commerce sites to identify common complaints and areas for improvement (Experiment 10).
- **Competitor Analysis** – Scraping competitor social media pages or review sites to benchmark performance
- **Content Creation** – Gathering popular quotes or discussions to inspire promotional content.

STEPS TO IMPLEMENT THE DATA COLLECTION PROCESS:

The following steps were followed to collect data from a website using Python.

Step 1: Install and Import Required Libraries

The required Python libraries were installed using the command prompt or terminal. The installation command installs the Requests library and BeautifulSoup library which are required for sending web requests and parsing HTML content. Open **Command Prompt / Terminal** and run: `pip install requests beautifulsoup4`



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```
[2]
✓ 55 # Install requests (for connection), BeautifulSoup4 (for parsing), and pandas (for data tables)
!pip install requests beautifulsoup4 pandas

...
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (2.32.4)
Requirement already satisfied: BeautifulSoup4 in /usr/local/lib/python3.12/dist-packages (4.13.5)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests) (2026.1.4)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from BeautifulSoup4) (2.8.3)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from BeautifulSoup4) (4.15.0)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

Step 1: Importing Libraries

```
import requests
from bs4 import BeautifulSoup
import pandas as pd
import matplotlib.pyplot as plt
```

Step 2: Choose Website

A sample website designed for scraping practice was selected. The website used for this experiment is:

<https://quotes.toscrape.com>

This website is safe for educational scraping and contains quotes that can be easily extracted.

```
# Step 2: Choose Website
url = "https://quotes.toscrape.com"
print(f"--- Target URL: {url} ---")

--- Target URL: https://quotes.toscrape.com ---
```

Step 3: Establish Connection & Check Status Code

```
# Step 3: Establish Connection & Check Status Code
print("\n[Step 3] Connecting to website...")
response = requests.get(url)

if response.status_code == 200:
    print(f"Success! Status Code: {response.status_code}")
else:
    print(f"Failed. Status Code: {response.status_code}")
    # Stop execution if connection fails by raising an error
    response.raise_for_status()

...
[Step 3] Connecting to website...
Success! Status Code: 200
```



Step 4: Parse HTML Content

```
# Step 4: Parse HTML Content
print("\n[Step 4] Parsing HTML content...")
soup = BeautifulSoup(response.text, "html.parser")
```

```
[Step 4] Parsing HTML content...
```

Step 5: Extract Data

```
▶ # Step 5: Extract Data
print("\n[Step 5] Extracting quotes and authors...")

# Finding all quote and author elements
quotes_elements = soup.find_all("span", class_="text")
author_elements = soup.find_all("small", class_="author")

data_list = []

# Loop through extracted elements and store them
for i in range(len(quotes_elements)):
    quote_text = quotes_elements[i].text
    author_name = author_elements[i].text

    data_list.append({
        "Author": author_name,
        "Quote": quote_text
    })

print(f"Successfully extracted {len(data_list)} quotes.")
```

...

```
[Step 5] Extracting quotes and authors...
Successfully extracted 10 quotes.
```

Step 6: Display Data

```
▶ # Step 6: Display Data
print("\n[Step 6] Creating Data Table and Displaying Data...")
df = pd.DataFrame(data_list)

# Display the first 5 rows
print(df.head())
```

...

```
[Step 6] Creating Data Table and Displaying Data...
      Author      Quote
0  Albert Einstein  "The world as we have created it is a process ...
1      J.K. Rowling  "It is our choices, Harry, that show what we t...
2  Albert Einstein  "There are only two ways to live your life. On...
3     Jane Austen   "The person, be it gentleman or lady, who has ...
4  Marilyn Monroe  "Imperfection is beauty, madness is genius and...
```




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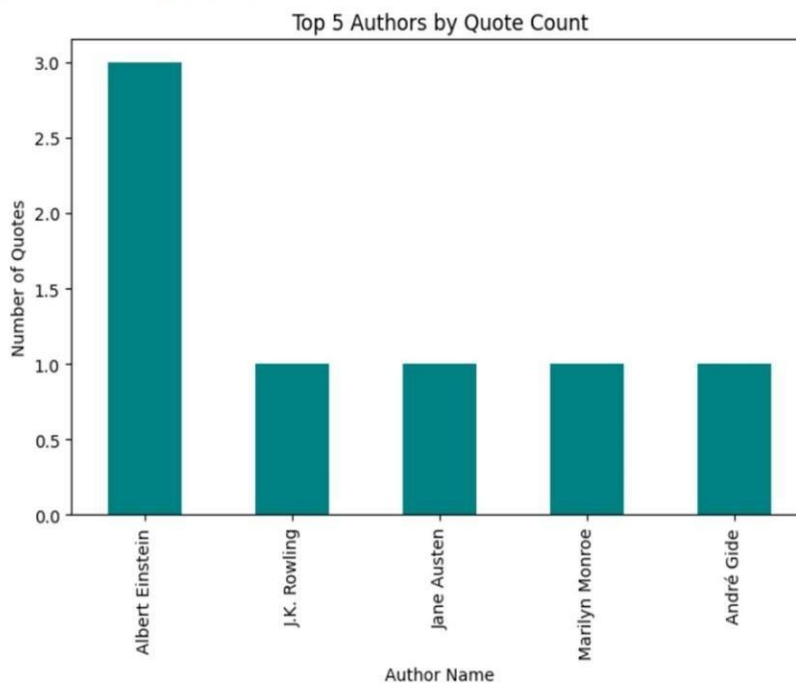
Step 7: Visualization

```
# Step 7: Visualization (Uniqueness)
print("\n[Step 7] Generating Graph...")

# Count how many quotes each author has
author_counts = df['Author'].value_counts().head(5)

# Plotting
plt.figure(figsize=(8, 5))
author_counts.plot(kind='bar', color='teal')
plt.title("Top 5 Authors by Quote Count")
plt.xlabel("Author Name")
plt.ylabel("Number of Quotes")
plt.show()
```

... [Step 7] Generating Graph...



CONCLUSION:

Therefore, we successfully implemented the process of collecting data from a website using web scraping techniques. The Python Libraries Requests and BeautifulSoup were used to connect to a webpage, retrieve HTML content, parse the structure, and extract textual information. This technique is widely used in social media analytics to gather large amounts of online data that can later be analyzed for business insights, sentiment analysis, and trend identification.